

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES							
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.						
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.						
20		VAT					***** MODULAR VALVES *****										
30																	
30		VA5	MODULAR DBB	K25RL10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN						
40			MODULAR DBB	K25RL16	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.						
50		VAV, 817	MODULAR DBB	K25CL16	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10						
60	U04	VAW	MODULAR DBB	M25RL10	0.75	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22						
70			MODULAR DBB	M25RL16	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - CONTROLLED BORE TYPE 2BVR	0	0	BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.						
80	U04	VAW	MODULAR DBB	M25HL10	0.75	1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	BA2	TO BE USED WITH RING TYPE JOINT FLANGES ONLY.						
90	U04	VAW	MODULAR DBB	M25HL10	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.						
100		VAU, 817	MODULAR DBB	M25YL10	1	1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
110		VAV, 817	MODULAR DBB	K25YL16	2	2	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL GO REDUCING - TYPE 2BV	3	3	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
120	U04	817	MODULAR DBB	M25CL10	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	UDD	USE IN LTCS SMALL BORE LINES ONLY						
130	U04	VAU, 817	MODULAR DBB	M25YL10L	1	1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	UDE	USE FOR CONECTION TO EXTRUDED BRANCH FITTINGS ONLY						
140	U04	VAV, 817	MODULAR DBB	K25NL16	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF X RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS						
150	U04	VAV, 817	MODULAR DBB	K25NL16L	3	3	MODULAR VALVE DB&B BALL TYPE 2500# CF X RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	VAT	CONTROLLED BORE (2BVR) TYPE VALVE TO BE USED FOR INSTRUMENT ISOLATION ONLY						
160	U04	VAW	MODULAR DBB	M25HL10L	0.75	1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END						
170	U04	VAW	MODULAR DBB	M25HL10L	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.						
180		VB2					***** BALL VALVES *****			VAV	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.						
180																	
190																	
190			BALL	A25RL15	2	2	BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO	0	0								
200	U04	817	BALL	A25CL15	3	3	BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO	0	0								
210	U04	817	BALL	A25CH15	6	14	BALL 2500# CF F22 INCONEL 625 TRUNNION SOFT RB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.						
220			BALL	B25RL09	0.75	1.5	BALL 2500# RTJ INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.						
230		VA5	BALL	B25RL15	2	2	BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO	0	0								
240	U04	817	BALL	B25CL15	3	3	BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO	0	0								
250	U04	817	BALL	B25CH15	6	14	BALL 2500# CF F22 INCONEL 625 TRIM TRUNNION SOFT FB GO	0	0								
260																	
260							***** THROUGH CONDUIT GATE VALVES *****										
270																	
270	U04	817	GATE	W25CH70	2	2	GATE 2500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB HW	0	0								
280	U04	817	GATE	W25CH71	3	12	GATE 2500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB GO	0	0								
290																	
290							***** GLOBE VALVES *****										
300																	
300			GLOBE	L25RL43	0.75	2	GLOBE 2500# RTJ INC 625 B564 INC 625 TRIM OS&Y HW	0	0								
310	U04	817	GLOBE	L25CL44	3	3	GLOBE 2500# CF INC 625 B564 INC 625 TRIM OS&Y GO	0	0								
320																	
320							***** CHECK VALVES *****										
330																	
330		VC1	CHECK	C25RL51	0.75	1.5	CHECK PISTON 2500# RTJ INC 625 TRIM 625	0	0								
340	U04		CHECK	C25RL53	2	2	CHECK DUAL 2500# RTJ INC 625 TRIM 625	0	0								
350	U04	817	CHECK	C25CL53	3	3	CHECK DUAL DBL FLG 2500# CF INC 625 TRIM 625	0	0								
360	U04	817	CHECK	C25CH53	6	24	CHECK DUAL DBL FLG 2500# CF F22 TRIM 625	0	0								
370																	
370							***** PIPE *****										
380																	
380		812	PIPE	5358511	0.75	1	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0								
390			PIPE	5358523	1.5	1.5	PIPE WT 11.0MM SMLS LTCS A333 GR.6 PE	0	0								
400			PIPE	5360061	2	2	PIPE WT 9.0MM SMLS 2-1/4CR F22M BE	0	0								
410			PIPE	5360064	3	3	PIPE WT 11.5MM SMLS 2-1/4CR F22M BE	0	0								
420			PIPE	5360067	4	4	PIPE WT 14.0MM SMLS 2-1/4CR F22M BE	0	0								
430			PIPE	5360068	6	6	PIPE WT 19.5MM SMLS 2-1/4CR F22M BE	0	0								
440			PIPE	5360072	8	8	PIPE WT 24.0MM SMLS 2-1/4CR F22M BE	0	0								
450			PIPE	5360074	10	10	PIPE WT 29.0MM SMLS 2-1/4CR F22M BE	0	0								
460			PIPE	5360078	12	12	PIPE WT 34.0MM SMLS 2-1/4CR F22M BE	0	0								
			TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
					ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO		MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -40 / 180 °C		12/16/2024	BY OZAC	U04	Page 1 of 4	2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
470			PIPE	5360081	14	14	PIPE WT 37.0MM SMLS 2-1/4CR F22M BE	0	0		
480			PIPE	5360084	16	16	PIPE WT 42.0MM SMLS 2-1/4CR F22M BE	0	0		
490			PIPE	5360086	18	18	PIPE WT 46.5MM SMLS 2-1/4CR F22M BE	0	0		
500			PIPE	5360087	20	20	PIPE WT 51.5MM SMLS 2-1/4CR F22M BE	0	0		
510			PIPE	5360094	26	26	PIPE WT 66.0MM SMLS 2-1/4CR F22M BE	0	0		
520											
520							***** FITTINGS *****				
530											
530			ELBOW	5392870	0.75	1	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
540			ELBOW	5392890	1.5	1.5	ELL 90 DEG LR WT 11.0MM LTCS A420 WPL6	0	0		
550			ELBOW	5406623	2	2	ELL 90 DEG LR WT 9.0MM 2-1/4CR F22M	0	0		
560			ELBOW	5406624	3	3	ELL 90 DEG LR WT 11.5MM 2-1/4CR F22M	0	0		
570			ELBOW	5406637	4	4	ELL 90 DEG LR WT 14.0MM 2-1/4CR F22M	0	0		
580			ELBOW	5406625	6	6	ELL 90 DEG LR WT 19.5MM 2-1/4CR F22M	0	0		
590			ELBOW	5406602	8	8	ELL 90 DEG LR WT 24.0MM 2-1/4CR F22M	0	0		
600			ELBOW	5406604	10	10	ELL 90 DEG LR WT 29.0MM 2-1/4CR F22M	0	0		
610			ELBOW	5406608	12	12	ELL 90 DEG LR WT 34.0MM 2-1/4CR F22M	0	0		
620			ELBOW	5406611	14	14	ELL 90 DEG LR WT 37.0MM 2-1/4CR F22M	0	0		
630			ELBOW	5406614	16	16	ELL 90 DEG LR WT 42.0MM 2-1/4CR F22M	0	0		
640			ELBOW	5406616	18	18	ELL 90 DEG LR WT 46.5MM 2-1/4CR F22M	0	0		
650			ELBOW	5406617	20	20	ELL 90 DEG LR WT 51.5MM 2-1/4CR F22M	0	0		
660			ELBOW	5406634	26	26	ELL 90 DEG LR WT 66.0MM 2-1/4CR F22M	0	0		
670			ELBOW	5392876	0.75	1	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
680			ELBOW	5392891	1.5	1.5	ELL 45 DEG LR WT 11.0MM LTCS A420 WPL6	0	0		
690			ELBOW	5407085	2	2	ELL 45 DEG LR WT 9.0MM 2-1/4CR F22M	0	0		
700			ELBOW	5407086	3	3	ELL 45 DEG LR WT 11.5MM 2-1/4CR F22M	0	0		
710			ELBOW	5407084	4	4	ELL 45 DEG LR WT 14.0MM 2-1/4CR F22M	0	0		
720			ELBOW	5407087	6	6	ELL 45 DEG LR WT 19.5MM 2-1/4CR F22M	0	0		
730			ELBOW	5407062	8	8	ELL 45 DEG LR WT 24.0MM 2-1/4CR F22M	0	0		
740			ELBOW	5407068	12	12	ELL 45 DEG LR WT 34.0MM 2-1/4CR F22M	0	0		
750			ELBOW	5407071	14	14	ELL 45 DEG LR WT 37.0MM 2-1/4CR F22M	0	0		
760			ELBOW	5407074	16	16	ELL 45 DEG LR WT 42.0MM 2-1/4CR F22M	0	0		
770			ELBOW	5407076	18	18	ELL 45 DEG LR WT 46.5MM 2-1/4CR F22M	0	0		
780			ELBOW	5407077	20	20	ELL 45 DEG LR WT 51.5MM 2-1/4CR F22M	0	0		
790			ELBOW	5407094	26	26	ELL 45 DEG LR WT 66.0MM 2-1/4CR F22M	0	0		
800											
800			TEE	5392879	0.75	1	TEE SCH XXS LTCS A420 WPL6	0	0		
810			TEE	5392892	1.5	1.5	TEE WT 11.0MM LTCS A420 WPL6	0	0		
820			TEE	5407182	2	2	TEE WT 9.0MM 2-1/4CR F22M	0	0		
830			TEE	5407183	3	3	TEE WT 11.5MM 2-1/4CR F22M	0	0		
840			TEE	5407181	4	4	TEE WT 14.0MM 2-1/4CR F22M	0	0		
850			TEE	5407184	6	6	TEE WT 19.5MM 2-1/4CR F22M	0	0		
860			TEE	5407162	8	8	TEE WT 24.0MM 2-1/4CR F22M	0	0		
870			TEE	5407168	12	12	TEE WT 34.0MM 2-1/4CR F22M	0	0		
880			TEE	5407170	14	14	TEE WT 37.0MM 2-1/4CR F22M	0	0		
890			TEE	5407173	16	16	TEE WT 42.0MM 2-1/4CR F22M	0	0		
900			TEE	5407175	18	18	TEE WT 46.5MM 2-1/4CR F22M	0	0		
910			TEE	5407176	20	20	TEE WT 51.5MM 2-1/4CR F22M	0	0		
920			TEE	5407212	26	26	TEE WT 66.0MM 2-1/4CR F22M	0	0		
930			TEE RED	5392877	0.75	0.75	TEE RED SCH XXS LTCS A420 WPL6	1	1		
940			TEE RED	5392894	0.75	1	TEE RED WT 11.0MM / SCH XXS LTCS A420 WPL6	1.5	1.5		
950			TEE RED	5407397	0.75	1	TEE RED BW WT 9.0MM / XXS 2-1/4CR F22M	2	2		
960			TEE RED	5407398	1.5	1.5	TEE RED BW WT 9.0MM / 11.0MM 2-1/4CR F22M	2	2		
970			TEE RED	5407399	1.5	1.5	TEE RED BW WT 11.5MM / 11.0MM 2-1/4CR F22M	3	3		
980			TEE RED	5407400	2	2	TEE RED BW WT 11.5MM / 9.0MM 2-1/4CR F22M	3	3		
990	U04		TEE RED	5407401	2	2	TEE RED BW WT 14.0MM / 9.0MM 2-1/4CR F22M	4	4		
1000			TEE RED	5407409	8	8	TEE RED BW WT 51.5MM / 24.0MM 2-1/4CR F22M	20	20		
1010			TEE RED	5407411	10	10	TEE RED BW WT 37.0MM / 29.0MM 2-1/4CR F22M	14	14		
1020			TEE RED	5407404	10	10	TEE RED BW WT 46.5MM / 29.0MM 2-1/4CR F22M	18	18		
1030			TEE RED	5407410	12	12	TEE RED BW WT 46.5MM / 34.0MM 2-1/4CR F22M	18	18		
1040			TEE RED	5407405	16	16	TEE RED BW WT 46.5MM / 42.0MM 2-1/4CR F22M	18	18		
1050			TEE RED	5407406	16	16	TEE RED BW WT 51.5MM / 42.0MM 2-1/4CR F22M	20	20		
1060			TEE RED	5407407	18	18	TEE RED BW WT 51.5MM / 46.5MM 2-1/4CR F22M	20	20		
1070			TEE RED	5407408	20	20	TEE RED BW WT 66.0MM / 51.5MM 2-1/4CR F22M	26	26		
1080											
1080	U04		CAP	5392880	0.75	1	CAP BW SCH XXS LTCS A420 WPL6	0	0		
1090			CAP	5392893	1.5	1.5	CAP WT 11.0MM LTCS A420 WPL6	0	0		
1100			CAP	5407126	2	2	CAP WT 9.0MM 2-1/4CR F22M	0	0		
1110			CAP	5407127	3	3	CAP WT 11.5MM 2-1/4CR F22M	0	0		
1120			CAP	5407125	4	4	CAP WT 14.0MM 2-1/4CR F22M	0	0		
1130			CAP	5407128	6	6	CAP WT 19.5MM 2-1/4CR F22M	0	0		
1140			CAP	5407102	8	8	CAP WT 24.0MM SMLS 2-1/4CR F22M	0	0		
1150			CAP	5407108	12	12	CAP WT 34.0MM SMLS 2-1/4CR F22M	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -40 / 180 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1160			CAP	5407111	14	14	CAP WT 37.0MM SMLS 2-1/4CR F22M	0	0	
1170			CAP	5407114	16	16	CAP WT 42.0MM SMLS 2-1/4CR F22M	0	0	
1180			CAP	5407116	18	18	CAP WT 46.5MM 2-1/4CR F22M	0	0	
1190			CAP	5407117	20	20	CAP WT 51.5MM 2-1/4CR F22M	0	0	
1200			CAP	5407123	26	26	CAP WT 66.0MM 2-1/4CR F22M	0	0	
1210										
1210			RED CONC	5392881	0.75	0.75	REDUCER CONC SCH XXS LTCS A420 WPL6	1	1	
1220			RED CONC	5392895	0.75	1	REDUCER CONC WT 11.0MM / SCH XXS LTCS A420 WPL6	1.5	1.5	
1230			RED CONC	5407246	0.75	1	REDUCER CONC WT 9.0MM / XXS 2-1/4CR F22M	2	2	
1240			RED CONC	5407247	1.5	1.5	REDUCER CONC WT 9.0MM / 11.0MM 2-1/4CR F22M	2	2	
1250			RED CONC	5407248	1.5	1.5	REDUCER CONC WT 11.5MM / 11.0MM 2-1/4CR F22M	3	3	
1260			RED CONC	5407249	1.5	1.5	REDUCER CONC WT 14.0MM / 11.0MM 2-1/4CR F22M	4	4	
1270			RED CONC	5407250	2	2	REDUCER CONC WT 11.5MM / 9.0MM 2-1/4CR F22M	3	3	
1280			RED CONC	5407251	2	2	REDUCER CONC WT 14.0MM / 9.0MM 2-1/4CR F22M	4	4	
1290			RED CONC	5407252	3	3	REDUCER CONC WT 14.0MM / 11.5MM 2-1/4CR F22M	4	4	
1300			RED CONC	5407293	8	8	REDUCER CONC WT 29.0MM/24.0MM 2-1/4CR F22M	10	10	
1310			RED CONC	5407295	10	10	REDUCER CONC WT 37.0MM/ 29.0MM 2-1/4CR F22M	14	14	
1320			RED CONC	5407294	12	12	REDUCER CONC WT 51.5MM/34.0MM 2-1/4CR F22M	20	20	
1330			RED CONC	5407296	16	16	REDUCER CONC WT 46.5MM/ 42.0MM 2-1/4CR F22M	18	18	
1340			RED CONC	5407297	16	16	REDUCER CONC WT 51.5MM/42.0MM 2-1/4CR F22M	20	20	
1350			RED CONC	5407299	18	18	REDUCER CONC WT 51.5MM/46.5MM 2-1/4CR F22M	20	20	
1360			RED CONC	5407277	20	20	REDUCER CONC WT 66.0MM/51.5MM 2-1/4CR F22M	26	26	
1370			RED ECC	5392882	0.75	0.75	REDUCER ECC SCH XXS LTCS A420 WPL6	1	1	
1380			RED ECC	5392896	0.75	1	REDUCER ECC WT 11.0MM / SCH XXS LTCS A420 WPL6	1.5	1.5	
1390			RED ECC	5407305	0.75	1	REDUCER ECC WT 9.0MM / XXS 2-1/4CR F22M	2	2	
1400			RED ECC	5407306	1.5	1.5	REDUCER ECC WT 9.0MM / 11.0MM 2-1/4CR F22M	2	2	
1410			RED ECC	5407307	1.5	1.5	REDUCER ECC WT 11.5MM / 11.0MM 2-1/4CR F22M	3	3	
1420			RED ECC	5407308	1.5	1.5	REDUCER ECC WT 14.0MM / 11.0MM 2-1/4CR F22M	4	4	
1430			RED ECC	5407309	2	2	REDUCER ECC WT 11.5MM / 9.0MM 2-1/4CR F22M	3	3	
1440			RED ECC	5407341	2	2	REDUCER ECC WT 14.0MM / 9.0MM 2-1/4CR F22M	4	4	
1450			RED ECC	5407342	3	3	REDUCER ECC WT 14.0MM / 11.5MM 2-1/4CR F22M	4	4	
1460			RED ECC	5407322	6	6	REDUCER ECC WT 34.0MM/ 19.5MM 2-1/4CR F22M	12	12	
1470			RED ECC	5407329	6	6	REDUCER ECC WT 24.0MM/ 19.5MM 2-1/4CR F22M	8	8	
1480			RED ECC	5407339	8	8	REDUCER ECC WT 29.0MM/ 24.0MM 2-1/4CR F22M	10	10	
1490			RED ECC	5407334	8	8	REDUCER ECC WT 34.0MM/ 24.0MM 2-1/4CR F22M	12	12	
1500			RED ECC	5407353	8	8	REDUCER ECC WT 37.0MM/ 24.0MM 2-1/4CR F22M	14	14	
1510			RED ECC	5407354	10	10	REDUCER ECC WT 37.0MM/ 29.0MM 2-1/4CR F22M	14	14	
1520			RED ECC	5407358	8	8	REDUCER ECC WT 42.0MM/ 24.0MM 2-1/4CR F22M	16	16	
1530			RED ECC	5407332	10	10	REDUCER ECC WT 34.0MM/ 29.0MM 2-1/4CR F22M	12	12	
1540			RED ECC	5407355	10	10	REDUCER ECC WT 42.0MM/ 29.0MM 2-1/4CR F22M	16	16	
1550			RED ECC	5407356	16	16	REDUCER ECC WT 46.5MM/ 42.0MM 2-1/4CR F22M	18	18	
1560			RED ECC	5407357	16	16	REDUCER ECC WT 51.5MM/ 42.0MM 2-1/4CR F22M	20	20	
1570			RED ECC	5407359	18	18	REDUCER ECC WT 51.5MM/ 46.5MM 2-1/4CR F22M	20	20	
1580			RED ECC	5407337	20	20	REDUCER ECC WT 66.0MM/ 51.5MM 2-1/4CR F22M	26	26	
1590							*****EXTRUDED BRANCHES*****			
1600							***** FLANGES *****			
1610		816								
1610			FLANGE	5563396	3	3	FLG WN 2500# RF 2-1/4CR F22M WT 11.5MM	0	0	
1620			FLANGE	5563397	4	4	FLG WN 2500# RF 2-1/4CR F22M WT 14.0MM	0	0	
1630			FLANGE	5563398	6	6	FLG WN 2500# RF 2-1/4CR F22M WT 19.5MM	0	0	
1640			FLANGE	5563400	8	8	FLG WN 2500# RF 2-1/4CR F22M WT 24.0MM	0	0	
1650			FLANGE	5563401	10	10	FLG WN 2500# RF 2-1/4CR F22M WT 29.0MM	0	0	
1660			FLANGE	5563402	12	12	FLG WN 2500# RF 2-1/4CR F22M WT 34.0MM	0	0	
1670			FLANGE	5563403	14	14	FLG WN 2500# RF 2-1/4CR F22M WT 37.0MM MANUFACTURERS STANDARD	0	0	
1680			FLANGE	5563404	16	16	FLG WN 2500# RF 2-1/4CR F22M WT 42.0MM MANUFACTURERS STANDARD	0	0	
1690			FLANGE	5563405	18	18	FLG WN 2500# RF 2-1/4CR F22M WT 46.5MM MANUFACTURERS STANDARD	0	0	
1700			FLANGE	5563406	20	20	FLG WN 2500# RF 2-1/4CR F22M WT 51.5MM MANUFACTURERS STANDARD	0	0	
1710			FLANGE	5563430	26	26	FLG WN 2500# RF 2-1/4CR F22M WT 66.0MM MANUFACTURERS STANDARD	0	0	
1720	U04	FA8	TEE RED	5599713	0.5	0.5	FLG SCRD (FNPT) REDUCING 2500# RTJ LTCS A350 LF2 CL.1	0.75	0.75	
1730										
1730	U04		FLG ADAPTER	5851202	0.75	2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE	0	0	
1740										
1740		UDE	FLANGE	5563339	0.75	1	FLG WN 2500# RTJ 2-1/4CR F22M SCH XXS	0	0	
1750		UDE	FLANGE	5563340	1.5	1.5	FLG WN 2500# RTJ 2-1/4CR F22M WT 11.0MM	0	0	
1760	U04	UDD	FLANGE	5557706	0.5	1	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XXS	0	0	
1770	U04	UDD	FLANGE	5619935	1.5	1.5	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 WT 11.0MM	0	0	
1780			FLANGE	5619932	2	2	FLG WN 2500# RTJ 2-1/4CR F22M WT 9MM	0	0	
1790	U04	817	FLANGE	5619921	3	3	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 11.5MM	0	0	
1800	U04	817	FLANGE	5619922	4	4	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 14.0MM	0	0	
1810	U04	817	FLANGE	5619923	6	6	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 19.5MM	0	0	
1820	U04	817	FLANGE	5619853	8	8	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 24.0MM	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -40 / 180 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1830	U04	817	FLANGE	5619854	10	10	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 29.0MM	0	0	
1840	U04	817	FLANGE	5619855	12	12	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 34.0MM	0	0	
1850	U04	817	FLANGE	5619869	14	14	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 37.0MM	0	0	
1860	U04	817	FLANGE	5619856	16	16	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 42.0MM	0	0	
1870	U04	817	FLANGE	5619857	18	18	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 46.5MM	0	0	
1880	U04	817	FLANGE	5619858	20	20	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 51.5MM	0	0	
1890	U04	817	FLANGE	5619859	26	26	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 66.0MM	0	0	
1900	U04		FLANGE	5563313	3	3	FLG WN 2500# RTJ 2-1/4CR F22M WT 11.5MM	0	0	
1910			FLANGE	5563312	10	10	FLG WN 2500# RTJ 2-1/4CR F22M WT 29.0MM	0	0	
1920	U04		BLIND	5619736	0.75	1	FLG BLIND 2500# RTJ LTCS A350 LF2 CL.1	0	0	
1930			BLIND	5619737	1.5	3	FLG BLIND 2500# RTJ 2-1/4CR F22M	0	0	
1940	U04	817	BLIND	5619792	3	3	FLG COMPACT BLIND 2500# RJ 2-1/4CR F22M PIPE CLASS 2500F22 FOR NPS 3 INCH ONLY	0	0	
1950	U04	817	BLIND	5619791	4	26	FLG COMPACT BLIND 2500# RJ 2-1/4CR F22M PIPE CLASS 2500F22 FOR NPS 4 INCH & ABOVE	0	0	
1960			BLIND	5407493	3	12	FLG BLIND 2500# RF 2-1/4CR F22M	0	0	
1970			BLIND	5407494	14	26	FLG BLIND 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
1980	U04	91	FLANGE	5563414	2	2	FLG WN 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 42.3MM 2-1/4CR F22M	0	0	
1990	U04	91	FLANGE	5563419	8	8	FLG WN 10000 API 6A TYPE 6BX 7 1/16" NS (8" NB) RTJ BORE 160.1MM 2-1/4CR F22M	0	0	
2000							***** PIPE NIPPLES *****			
2010	U04		NIPPLE	5541181	0.75	1	NIPPLE SCH XXS 2-1/4CR F22M BBE 4" LG	0	0	
2020	U04		NIPPLE	5541185	1.5	1.5	NIPPLE WT 11.0MM 2-1/4CR F22M BBE 4" LG	0	0	
2030							***** GASKETS *****			
2040										
2040			GASKET	5672271	3	12	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2050			GASKET	5672272	14	26	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR MANUFACTURERS STANDARD	0	0	
2060	U04		GASKET	5674480	3	3	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2070			GASKET	5674480	0.5	2	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2080			GASKET	5674480	10	10	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2090			GASKET	5674486	8	8	GASKET 10000 API 6A BX 156 7 1/16" NS 316SS RING	0	0	
2100	U04	817	GASKET	5673511	3	3	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 2500F22 FOR NPS 3 INCH ONLY	0	0	
2110	U04	817	GASKET	5673510	4	26	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 2500F22 FOR NPS 4 INCH & ABOVE	0	0	
2120	U04		GASKET	5674471	2	2	GASKET 10000 API 6A BX 151 1 13/16" NS 316 SS RING	0	0	
2130	U04		GASKET	5674487	10	10	GASKET 10000 API 6A BX 157 9" NS 316SS RING	0	0	
2140	U04		GASKET	5674483	12	14	GASKET 10000 API 6A BX 158 11" NS 316SS RING	0	0	
2150	U04		GASKET	5674484	16	18	GASKET 10000 API 6A BX 159 13 5/8" NS 316SS RING	0	0	
2160							***** BOLTING *****			
2160		B1								
2170										
2170		BA2	BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2180	U04	BA5	BOLTING	5675778	6	99	STUD-BOLT C/W 2 COMPACT NUTS API-6A718	0	0	
2190		BA1	BOLTING	5675776	10	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
2200										
2200		SB6	BOLTING	5675675	12	26	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2210							***** BLINDS & SPACERS *****			
2220										
2220			SPEC BLIND	5407692	2	2	SPECTACLE BLIND RTJ 2500# 2-1/4 CR F22M	0	0	
2230	U04	817	SPEC BLIND	5407694	3	3	SPECTACLE BLIND COMPACT 2500# RJ 2-1/4 CR F22M LINE CLASS 2500F22 FOR NPS 3 INCH & BELOW	0	0	
2240										
2240	U04	817	PADDLE	5407805	8	26	PADDLE BLANK COMPACT 2500# RJ 2-1/4CR F22M FOR LINE CLASS 2500F22 NPS 4 INCH & ABOVE	0	0	
2250	U04	817	PADDLE	5407705	8	26	PADDLE SPACER COMPACT 2500# RJ 2-1/4CR F22M FOR LINE CLASS 2500F22 NPS 4 INCH & ABOVE	0	0	
2260										
2260	U04		SPEC BLIND	5407693	3	3	SPECTACLE BLIND 2500# RF 2-1/4 CR F22M	0	0	
2270										
2270			PADDLE	5407803	8	8	PADDLE BLANK 2500# RF 2-1/4CR F22M	0	0	
2280			PADDLE	5407803	12	12	PADDLE BLANK 2500# RF 2-1/4CR F22M	0	0	
2290			PADDLE	5407703	8	8	PADDLE SPACER 2500# RF 2-1/4CR F22M	0	0	
2300			PADDLE	5407703	12	12	PADDLE SPACER 2500# RF 2-1/4CR F22M	0	0	
2310			PADDLE	5407804	14	26	PADDLE BLANK 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
2320			PADDLE	5407704	14	26	PADDLE SPACER 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
2330	U04	817	PADDLE	5407806	3	3	PADDLE BLANK COMPACT 2500# RJ 2-1/4CR F22M FOR LINE CLASS 2500F22 NPS 3 INCH & BELOW	0	0	
2340	U04	817	PADDLE	5407706	3	3	PADDLE SPACER COMPACT 2500# RJ 2-1/4CR F22M FOR LINE CLASS 2500F22 NPS 3 INCH & BELOW	0	0	
2350										

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